

Notes and Solutions for *Semiparametric Theory and Missing Data*

A companion to Tsiatis (2006)

Lu Mao

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Home

This book develops explanatory notes, derivations, worked examples, and original solutions for exercises accompanying *Semiparametric Theory and Missing Data* by Anastasios A. Tsiatis.

The objective is to make the geometry of semiparametric models and the theory of missing and coarsened data easier to learn and apply. The presentation follows the organization of the source book while remaining independently written.

How to use the book

Each chapter contains the following components:

- Learning objectives
- A concise chapter summary
- Solutions to selected exercises from the source text
- Additional exercises

Source text

Readers should consult Tsiatis (2006) for the original statements, examples, and exercises. Exercise numbers in this companion refer to that text.

Preface

Purpose

These notes are intended for graduate students and researchers in biostatistics and statistics who want a detailed path through semiparametric theory, influence functions, missing-data methods, and causal inference.

Conventions

- Random variables are written in uppercase and realizations in lowercase
- Vectors are treated as column vectors unless stated otherwise
- Expectations and inner products are taken under the true data-generating distribution unless annotated
- A solution labeled “Exercise $j.k$ ” corresponds to that numbered exercise in Tsiatis (2006)

Scope and attribution

This is a companion volume, not a reproduction of the source. Explanations, derivations, examples, and solutions should be written independently. Page and exercise references identify where the corresponding topic appears in the source.

Notation

Symbol	Meaning
Z	Full observed random vector for one independent unit
$Z_1, \dots, Z_n$	Independent and identically distributed observations
β	Finite-dimensional parameter of interest
η	Nuisance parameter, often infinite-dimensional
$\langle h_1, h_2 \rangle = E\{h_1^\top(Z)h_2(Z)\}$	Inner product in the Hilbert space of mean-zero, square-integrable functions
S_β	Score for β
Λ	Nuisance tangent space
$\Lambda^\perp$	Orthogonal complement of a nuisance tangent space
φ	Influence function
φ^{eff}	Efficient influence function
R	Indicator that the full data are observed
$\pi(\cdot)$	Observation or response probability

I will add chapter-specific notation to this table as the manuscript develops.

Part I

I. Semiparametric foundations

1 Introduction to semiparametric models

1.1 Learning Objectives

- Distinguish parametric, nonparametric, and semiparametric models
- Identify parameters of interest and nuisance parameters
- Recognize restricted moment and proportional hazards models as semiparametric

1.2 Chapter Summary

A **parametric model** specifies the data distribution through a finite-dimensional parameter:

$$p(z; \theta), \quad \theta \in \mathbb{R}^p.$$

A **semiparametric model** retains a finite-dimensional parameter of interest, β , while leaving other aspects of the distribution unspecified through an infinite-dimensional nuisance parameter $\eta(\cdot)$:

$$p\{z; \beta, \eta(\cdot)\}, \quad \beta \in \mathbb{R}^q.$$

The nuisance parameter is often an unknown function, such as a conditional distribution, a covariate distribution, or a baseline hazard function. But it can also be defined *implicitly* as the “other” aspects of the distribution that are not of direct interest. This allows the model to target β without imposing a full parametric constraint on the data.

Key examples include:

- **Restricted moment model.** Only the conditional mean is specified:

$$E(Y \mid X) = \mu(X, \beta).$$

- Here η relates to aspects of the conditional distribution of $Y \mid X$ other than the mean, as well as the marginal distribution of X .

- **Proportional hazards model.** Covariate effects are modeled by β , while the baseline hazard remains unspecified:

$$\lambda(t \mid X) = \lambda_0(t) \exp(\beta^\top X).$$

- Here $\eta = \lambda_0(t)$.

- **Nonparametric model.** The distribution is unrestricted, and the parameter of interest is a functional of that distribution:

$$p_Z(z) = \eta(z), \quad \beta = E(Z) \quad \text{or} \quad E(Z^2).$$

A semiparametric estimator $\hat{\beta}_n$ is typically required to be consistent and asymptotically normal:

$$\hat{\beta}_n \xrightarrow{P} \beta, \quad n^{1/2}(\hat{\beta}_n - \beta) \xrightarrow{D} N(0, \Sigma).$$

The significance of a semiparametric estimator is that the above holds regardless of the underlying, and potentially complex, state of η .

The chapter motivates two key questions throughout the book:

- Constructing regular semiparametric estimators;
- Finding an optimal semiparametric estimator with the smallest asymptotic variance.

The answers rely on the geometry of influence functions, tangent spaces, orthogonality, and projection in Hilbert spaces.

1.3 Additional Problems

Problem 1.1

Classify each model as parametric, semiparametric, or nonparametric. Identify the parameter of interest and any nuisance parameter.

1. $Y \sim N(\mu, \sigma^2)$, with interest in μ .
2. $E(Y | X) = \beta_0 + \beta_1 X$.
3. $p_Z(z)$ is unrestricted, with interest in $E(Z)$

Problem 1.2

Suppose that

$$E(Y | X) = \beta^\top X,$$

where X is a q -dimensional random vector and $E(XX^\top)$ is nonsingular.

1. Show that β satisfies

$$E[X\{Y - \beta^\top X\}] = 0.$$

2. Show that

$$\beta = \{E(XX^\top)\}^{-1}E(XY).$$

3. Propose an estimator of β based on an iid sample (Y_i, X_i) , $i = 1, \dots, n$.
4. Explain why this estimator does not require a parametric model for $Y \mid X$.

2 Hilbert space for random vectors

2.1 Learning Objectives

- Work with mean-zero, square-integrable random functions
- Interpret orthogonality through covariance
- Use the projection theorem in statistical problems

2.2 Chapter Summary

This chapter introduces the Hilbert space of q -dimensional random functions $h(Z)$ with mean zero and finite second moments:

$$E\{h(Z)\} = 0, \quad E\{h^\top(Z)h(Z)\} < \infty.$$

The covariance inner product is defined by

$$\langle h_1, h_2 \rangle = E(h_1^\top h_2),$$

with corresponding norm

$$\|h\| = \langle h, h \rangle^{1/2}.$$

Two elements are **orthogonal** when their inner product is zero. For a closed linear subspace $\mathcal{U} \subset \mathcal{H}$, the projection theorem guarantees a unique projection $\Pi(h \mid \mathcal{U}) \in \mathcal{U}$ such that

$$\Pi(h \mid \mathcal{U}) = \arg \min_{u \in \mathcal{U}} \|h - u\|.$$

The residual after projection is orthogonal to the subspace:

$$h - \Pi(h \mid \mathcal{U}) \perp \mathcal{U}.$$

For a finite-dimensional subspace spanned by a random vector $u(Z)$, the projection is a linear function of u , whose coefficients are determined by the variance of $u(Z)$ and its covariance with $h(Z)$ (see (2.1) below).

2.3 Notes and Extensions

2.3.1 Projection onto linear span vs conditional expectation

By definition,

$$\Pi\{h(Z) \mid \mathcal{U}\} = \arg \min_{g \in \mathcal{U}} \|h - g\|,$$

for any closed linear subspace $\mathcal{U} \subset \mathcal{H}$. That is, the projection of $h(Z)$ onto $\mathcal{U}$ is the element of $\mathcal{U}$ closest to $h(Z)$ in the norm $\|\cdot\|$.

Example 1 in Tsiatis (2006) considers the linear span

$$\mathcal{U} = \{a^\top u : a \in \mathbb{R}^k\}$$

of a k -dimensional random vector

$$u(Z) = \{u_1(Z), \dots, u_k(Z)\}^\top.$$

It establishes that

$$\Pi\{h(Z) \mid \mathcal{U}\} = E(hu^\top) \{E(uu^\top)\}^{-1} u. \quad (2.1)$$

If $\mathcal{U}$ is expanded to include all measurable square-integrable functions of $u(Z)$, the projection is the conditional expectation of $h(Z)$ given $u(Z)$ (Tsiatis, 2006). Specifically,

$$E\{h(Z) \mid u\} = \Pi\{h(Z) \mid \mathcal{U}^*\} = \arg \min_{g \in \mathcal{U}^*} \|h - g\|,$$

where

$$\mathcal{U}^* = \{a(u) : a(\cdot) \text{ is a measurable and square-integrable function on } \mathbb{R}^k\}.$$

Because $\mathcal{U} \subset \mathcal{U}^*$, the best approximation of $h(Z)$ in $\mathcal{U}^*$ is at least as good as the best approximation in $\mathcal{U}$. Thus,

$$\|h - E\{h(Z) \mid u\}\| \leq \|h - \Pi\{h(Z) \mid \mathcal{U}\}\|.$$

By a standard property of the multivariate normal distribution, $E\{h(Z) \mid u\}$ is linear in u when $h(Z)$ and $u(Z)$ are jointly multivariate normal. In this case,

$$E\{h(Z) \mid u\} = \Pi\{h(Z) \mid \mathcal{U}\},$$

which is in turn given by (2.1).

2.3.2 Dimensionality of $\mathcal{U} = \{Bv : B \in \mathbb{R}^{q \times r}\}$

Here, v is an r -dimensional random vector. On page 17, it is claimed that $\mathcal{U}$ is $q \times r$ -dimensional because it is the linear span of that many linearly independent functions.

Indeed, a typical element of $\mathcal{U}$ is given by

$$u = Bv = \begin{pmatrix} \sum_{j=1}^r B_{1j}v_j \\ \sum_{j=1}^r B_{2j}v_j \\ \vdots \\ \sum_{j=1}^r B_{qj}v_j \end{pmatrix} = \sum_{j=1}^r \begin{pmatrix} B_{1j}v_j \\ B_{2j}v_j \\ \vdots \\ B_{qj}v_j \end{pmatrix} = \sum_{j=1}^r \sum_{i=1}^q B_{ij}u_{ij}^{q \times 1},$$

where

$$u_{ij}^{q \times 1} = (\underbrace{0, \dots, 0}_{i-1} \ v_j, \underbrace{0, \dots, 0}_{q-i})^\top,$$

as defined in the book.

2.4 Solutions to Exercises

2.4.1 Exercise 2.1

Solution. Let $\mathcal{U}$ be a closed linear subspace of the Hilbert space $\mathcal{H}$, and let $h \in \mathcal{H}$. We prove:

- **Existence:** There exists $u_0 \in \mathcal{U}$ such that

$$\|h - u_0\| = \inf_{u \in \mathcal{U}} \|h - u\|.$$

- **Uniqueness:** The minimizer u_0 is unique.
- **Orthogonality:** The residual is orthogonal to $\mathcal{U}$:

$$\langle h - u_0, u \rangle = 0 \quad \text{for every } u \in \mathcal{U}.$$

To show **existence**, let

$$d = \inf_{u \in \mathcal{U}} \|h - u\|.$$

Choose a sequence $\{u_n\} \subset \mathcal{U}$ such that

$$\|h - u_n\| \longrightarrow d.$$

We first show that $\{u_n\}$ is a Cauchy sequence. Because $(u_n + u_m)/2 \in \mathcal{U}$,

$$d \leq \left\| h - \frac{u_n + u_m}{2} \right\|.$$

Moreover,

$$\begin{aligned} \|u_n - u_m\|^2 &= \|(h - u_n) - (h - u_m)\|^2 \\ &= \|h - u_n\|^2 + \|h - u_m\|^2 - 2\langle h - u_n, h - u_m \rangle. \end{aligned}$$

Therefore,

$$\begin{aligned} 4d^2 &\leq 4 \left\| h - \frac{u_n + u_m}{2} \right\|^2 \\ &= \|(h - u_n) + (h - u_m)\|^2 \\ &= \|h - u_n\|^2 + \|h - u_m\|^2 + 2\langle h - u_n, h - u_m \rangle \\ &= 2\|h - u_n\|^2 + 2\|h - u_m\|^2 - \|u_n - u_m\|^2. \end{aligned}$$

Hence,

$$\begin{aligned} \|u_n - u_m\|^2 &\leq 2\|h - u_n\|^2 + 2\|h - u_m\|^2 - 4d^2 \\ &\longrightarrow 0 \quad \text{as } n, m \longrightarrow \infty. \end{aligned}$$

Thus, $\{u_n\}$ is Cauchy. Since $\mathcal{H}$ is complete, there exists $u_0 \in \mathcal{H}$ such that

$$u_n \longrightarrow u_0.$$

Because $\mathcal{U}$ is closed and $u_n \in \mathcal{U}$ for every n ,

$$u_0 \in \mathcal{U}.$$

By continuity of the norm,

$$\|h - u_0\| = \lim_{n \rightarrow \infty} \|h - u_n\| = d.$$

This establishes existence.

To show **orthogonality**, fix $u \in \mathcal{U}$ and define

$$g(\epsilon) = \|h - (u_0 - \epsilon u)\|^2.$$

Since $u_0 - \epsilon u \in \mathcal{U}$ for every $\epsilon \in \mathbb{R}$, $g(\epsilon)$ is minimized at $\epsilon = 0$. Hence,

$$\begin{aligned} 0 &= \left. \frac{\partial}{\partial \epsilon} g(\epsilon) \right|_{\epsilon=0} \\ &= 2\langle h - u_0, u \rangle. \end{aligned}$$

Therefore,

$$\langle h - u_0, u \rangle = 0 \quad \text{for every } u \in \mathcal{U}.$$

Finally, for **uniqueness**, suppose that u_0 and u_0^* both minimize $\|h - u\|$ over $\mathcal{U}$. Since $u_0 - u_0^* \in \mathcal{U}$, orthogonality gives

$$\|h - u_0^*\|^2 = \underbrace{\|h - u_0\|^2}_{d^2} + \underbrace{\|u_0 - u_0^*\|^2}_{\geq 0}.$$

Because $\|h - u_0^*\|^2 = d^2$, it follows that

$$\|u_0 - u_0^*\|^2 = 0,$$

and hence $u_0 = u_0^*$.

2.4.2 Exercise 2.2

Solution. This exercise is a direct application of Example 2 in the book.

(a) Because $\Sigma_{22} = E(Y_2 Y_2^\top)$ is nonsingular, the projection formula in Example 2 gives

$$\begin{aligned} \Pi(Y_1 \mid \mathcal{U}) &= E(Y_1 Y_2^\top) \{E(Y_2 Y_2^\top)\}^{-1} Y_2 \\ &= \Sigma_{12} \Sigma_{22}^{-1} Y_2, \end{aligned}$$

which is equal to $E(Y_1 \mid Y_2)$ (see Section 2.3.1).

(b) Thus, the residual is

$$Y_1 - \Pi(Y_1 | \mathcal{U}) = Y_1 - \Sigma_{12}\Sigma_{22}^{-1}Y_2.$$

Its squared norm is

$$\|Y_1 - \Pi(Y_1 | \mathcal{U})\|^2 = \text{tr}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}).$$

2.4.3 Exercise 2.3

Solution. Let

$$W = Z_1^2 - \sigma_1^2.$$

Then

$$\mathcal{U} = \{a_1W + a_2Z_2 : a_1, a_2 \in \mathbb{R}\}.$$

Because $(Z_1, Z_2)^\top$ is mean-zero multivariate normal, all third moments are zero. Thus,

$$E(Z_1W) = E(Z_1^3) - \sigma_1^2E(Z_1) = 0,$$

and

$$E(WZ_2) = E(Z_1^2Z_2) - \sigma_1^2E(Z_2) = 0,$$

where we have used

$$E(Z_1^2Z_2) = E\{Z_2 \underbrace{E(Z_1^2 | Z_2)}_{\sigma_1^2 - \sigma_2^{-2}\sigma_{12}^2}\} = (\sigma_1^2 - \sigma_2^{-2}\sigma_{12}^2)E(Z_2) = 0.$$

Thus, W is orthogonal to both Z_1 and Z_2 . This means that the problem essentially reduces to projecting Z_1 onto the linear span of Z_2 alone.

(a) The component of Z_1 in the direction of W is zero. Therefore,

$$\Pi(Z_1 | \mathcal{U}) = \frac{E(Z_1Z_2)}{E(Z_2^2)}Z_2 = \sigma_2^{-2}\sigma_{12}Z_2.$$

(b) The residual is

$$Z_1 - \Pi(Z_1 | \mathcal{U}) = Z_1 - \sigma_2^{-2}\sigma_{12}Z_2.$$

Its variance is

$$\text{var}\{Z_1 - \Pi(Z_1 | \mathcal{U})\} = \sigma_1^2 - \sigma_2^{-2}\sigma_{12}^2.$$

2.5 Additional Problems

Problem 2.1

Let $\mathcal{U}$ be a closed linear subspace of $\mathcal{H}$, and let

$$u_0 = \Pi(h \mid \mathcal{U}).$$

Show that, for every $u \in \mathcal{U}$,

$$\|h - u\|^2 = \|h - u_0\|^2 + \|u - u_0\|^2.$$

Problem 2.2

Let $Z \sim N(0, 1)$, and consider the Hilbert space of one-dimensional mean-zero functions of Z with finite variance. Let

$$\mathcal{U} = \text{span} \{Z, Z^2 - 1\}.$$

- (a) Show that Z and $Z^2 - 1$ are orthogonal.
- (b) Find $\Pi(Z^3 \mid \mathcal{U})$.
- (c) Compute the variance of

$$Z^3 - \Pi(Z^3 \mid \mathcal{U}).$$

Problem 2.3

Let $h_1, h_2 \in \mathcal{H}$. For any real number a , use

$$\|h_1 - ah_2\|^2 \geq 0$$

to prove the Cauchy–Schwarz inequality

$$|\langle h_1, h_2 \rangle|^2 \leq \|h_1\|^2 \|h_2\|^2.$$

Show that equality holds if and only if $h_1 = ah_2$ for some scalar a .

Problem 2.4

Let Z have support $\{z_1, z_2, z_3\}$, with positive probabilities π_1, π_2, π_3 .

- (a) Find two linearly independent mean-zero functions of Z that span the space of all one-dimensional mean-zero functions of Z .
- (b) What is the dimension of the space of all q -dimensional mean-zero functions of Z ?

3 The geometry of influence functions

3.1 Learning objectives

- Define regular asymptotically linear estimators
- Derive influence functions for M-estimators
- Characterize the efficient influence function geometrically

3.2 Chapter Summary

3.3 Notes and Extensions

3.4 Solutions to Exercises

3.4.1 Exercise 3.1

3.5 Additional Exercises

4 Semiparametric models

4.1 Learning objectives

- Construct parametric submodels
- Define nuisance tangent spaces
- Derive influence functions and efficiency bounds in semiparametric models

4.2 Chapter Summary

4.3 Notes and Extensions

4.4 Solutions to Exercises

4.4.1 Exercise 4.1

4.5 Additional Exercises

5 Other examples of semiparametric models

5.1 Learning objectives

- Analyze location-shift regression models
- Connect proportional hazards estimation to tangent-space geometry
- Derive efficient estimators for means and treatment contrasts

5.2 Chapter Summary

5.3 Notes and Extensions

5.4 Solutions to Exercises

5.4.1 Exercise 5.1

5.5 Additional Exercises

Part II

II. Missing and coarsened data

6 Models and methods for missing data

6.1 Learning objectives

- Distinguish MCAR, MAR, and nonignorable missingness
- Compare likelihood, imputation, and inverse-probability methods
- Explain the basis of doubly robust estimation

6.2 Chapter Summary

6.3 Notes and Extensions

6.4 Solutions to Exercises

6.4.1 Exercise 6.1

6.5 Additional Exercises

7 Missing and coarsening at random for semiparametric models

7.1 Learning objectives

- Formalize missing and coarsened data
- State coarsening-at-random assumptions
- Relate full-data and observed-data models

7.2 Chapter Summary

7.3 Notes and Extensions

7.4 Solutions to Exercises

7.4.1 Exercise 7.1

7.5 Additional Exercises

8 Inverse probability weighted complete-case estimators

8.1 Learning objectives

- Construct IPW complete-case estimating equations
- Establish consistency and asymptotic normality
- Estimate asymptotic variance when response probabilities are estimated

8.2 Chapter Summary

8.3 Notes and Extensions

8.4 Solutions to Exercises

8.4.1 Exercise 8.1

8.5 Additional Exercises

9 Semiparametric estimators with monotone coarsening

9.1 Learning objectives

- Represent monotone coarsening processes
- Derive semiparametric estimators under monotone missingness
- Connect censoring to monotone coarsening

9.2 Chapter Summary

9.3 Notes and Extensions

9.4 Solutions to Exercises

9.4.1 Exercise 9.1

9.5 Additional Exercises

Part III

III. Efficiency and robustness

10 Improving efficiency and double robustness with coarsened data

10.1 Learning objectives

- Project influence functions onto augmentation spaces
- Construct efficient augmented IPW estimators
- Explain double robustness under different coarsening patterns

10.2 Chapter Summary

10.3 Notes and Extensions

10.4 Solutions to Exercises

10.4.1 Exercise 10.1

10.5 Additional Exercises

11 Locally efficient estimators for coarsened-data semiparametric models

11.1 Learning objectives

- Derive observed-data efficient scores
- Compare likelihood and augmented-IPW representations
- Construct locally efficient estimators

11.2 Chapter Summary

11.3 Notes and Extensions

11.4 Solutions to Exercises

11.4.1 Exercise 11.1

11.5 Additional Exercises

12 Approximate methods for gaining efficiency

12.1 Learning objectives

- Define restricted classes of augmented estimators
- Derive optimal estimators within a restricted class
- Balance efficiency gains against computational feasibility

12.2 Chapter Summary

12.3 Notes and Extensions

12.4 Solutions to Exercises

12.4.1 Exercise 12.1

12.5 Additional Exercises

Part IV

IV. Causal inference and imputation

13 Doubly robust estimation of average causal treatment effects

13.1 Learning objectives

- State identification assumptions for point-treatment effects
- Express causal estimation as a coarsened-data problem
- Derive the augmented inverse-probability estimator for the ATE

13.2 Chapter Summary

13.3 Notes and Extensions

13.4 Solutions to Exercises

13.4.1 Exercise 13.1

13.5 Additional Exercises

14 Multiple imputation from a frequentist perspective

14.1 Learning objectives

- Compare full-data and observed-data information
- Derive the asymptotic distribution of multiple-imputation estimators
- Explain proper imputation and Rubin's variance estimator

14.2 Chapter Summary

14.3 Notes and Extensions

14.4 Solutions to Exercises

14.4.1 Exercise 14.1

14.5 Additional Exercises

References

Tsiatis, Anastasios A. (2006). Semiparametric theory and missing data. Springer. <https://doi.org/10.1007/0-387-37345-4>.

A Linear algebra and projections

This appendix will collect matrix identities, projection formulas, and block-matrix calculations used throughout the book.

A.1 Matrix differentiation

A.2 Block-matrix inversion

A.3 Orthogonal projections

B Probability and asymptotic tools

This appendix will collect conditional-expectation identities, convergence results, empirical-process notation, and asymptotic linearity arguments.

B.1 Conditional expectation

B.2 Modes of convergence

B.3 Central limit arguments

C R code and reproducibility

This appendix contains reusable R functions and reproducibility conventions for computational examples.

C.1 Session information

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 10 x64 (build 19045)
```

```
Matrix products: default
LAPACK version 3.12.1
```

```
locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8
```

```
time zone: America/Chicago
tzcode source: internal
```

```
attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

```
loaded via a namespace (and not attached):
[1] compiler_4.5.3    fastmap_1.2.0     cli_3.6.5         htmltools_0.5.9
[5] tools_4.5.3       otel_0.2.0        rstudioapi_0.18.0 yaml_2.3.12
[9] tinytex_0.58      rmarkdown_2.30    knitr_1.51        jsonlite_2.0.0
[13] xfun_0.56         digest_0.6.39     rlang_1.1.7       evaluate_1.0.5
```